FUNCTIONAL TREAT

Data Analyzer and Transformer – Python Project

Project Output & Demonstration Report

This report documents the implemented menu-driven program and shows representative output for the sample dataset.

Sample dataset

1D: 34, 12, 56, 78, 43, 21, 90

2D: [[1, 2, 3], [4, 5, 6]]

Expected analysis

Metric	Result
Total elements	7
Minimum	12
Maximum	90
Sum	334
Average	47.71
Filtered (>= 50)	[56, 78, 90]
Sorted ascending	[12, 21, 34, 43, 56, 78, 90]
Factorial (5)	120

Features implemented

Built-in functions (len, sum, min, max); user-defined functions; *args and **kwargs; docstrings; recursive factorial and Fibonacci; lambda with filter() and map(); global dataset summary; multiple return values; 1D/2D lists; and sort() versus sorted().

Sample console execution

The following transcript is generated by running the submitted Python source with sample inputs. It demonstrates the menu workflow and expected calculations.

```
Welcome to Functional Treat!
Please enter your choice:
Choose input type:
Your choice: Enter numbers separated by spaces: Data stored successfully.
Current dataset:
Please enter your choice:
Current dataset:
Data Summary (built-in functions):
- Total elements: 7
- Minimum: 12
- Maximum: 90
- Sum: 334
- Average: 47.71
- Average via *args function: 47.71
Please enter your choice: Enter F for factorial or B for Fibonacci: Enter a non-negative integer: Factorial of 5 = 120
Please enter your choice: Enter F for factorial or B for Fibonacci: Enter a non-negative integer: First 7 Fibonacci terms: [0,
Please enter your choice: Enter threshold: Keep values A)bove/equal or B)elow/equal? Filtered values: [56, 78, 90]
Squared values using map() + lambda: [1156, 144, 3136, 6084, 1849, 441, 8100]
Please enter your choice: A)scending or D)escending? I)n-place sort() or N)ew sorted() list? Sorted values: [12, 21, 34, 43, 5
Please enter your choice:
Dataset Statistics:
- Minimum: 12
- Maximum: 90
- Sum: 334
- Average: 47.71
Please enter your choice:
Dataset summary (**kwargs):
- Minimum: 12
- Maximum: 90
- Average: 47.714285714285715
Global summary: {'count': 7, 'total': 334, 'average': 47.714285714285715}
Please enter your choice:
Choose input type:
Your choice: Enter rows separated by ';' and values by spaces: Data stored successfully.
Current dataset:
Please enter your choice: Flattened 2D data: [1, 2, 3, 4, 5, 6]
Please enter your choice: Thank you for using Functional Treat. Goodbye!
```

How to run the project

Open a terminal in the project folder and execute `python functional_treat.py`. Choose option 1 to enter data, then select any analysis or transformation from the menu. See README.md for full instructions and assumptions.

Note

This is a sample output report, not a screenshot of a live terminal. Results are based on the sample values stated in this document. Run the program to explore other inputs.